The Thoracic wall

Objectives

- The thoracic cage
- The intercostal space and muscles
- Blood vessels
- Nerves
- Lymphatic drainage of thoracic wall

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The thoracic wall

1- Thoracic cage :

Ribs

Sternum

Thoracic vertebrae

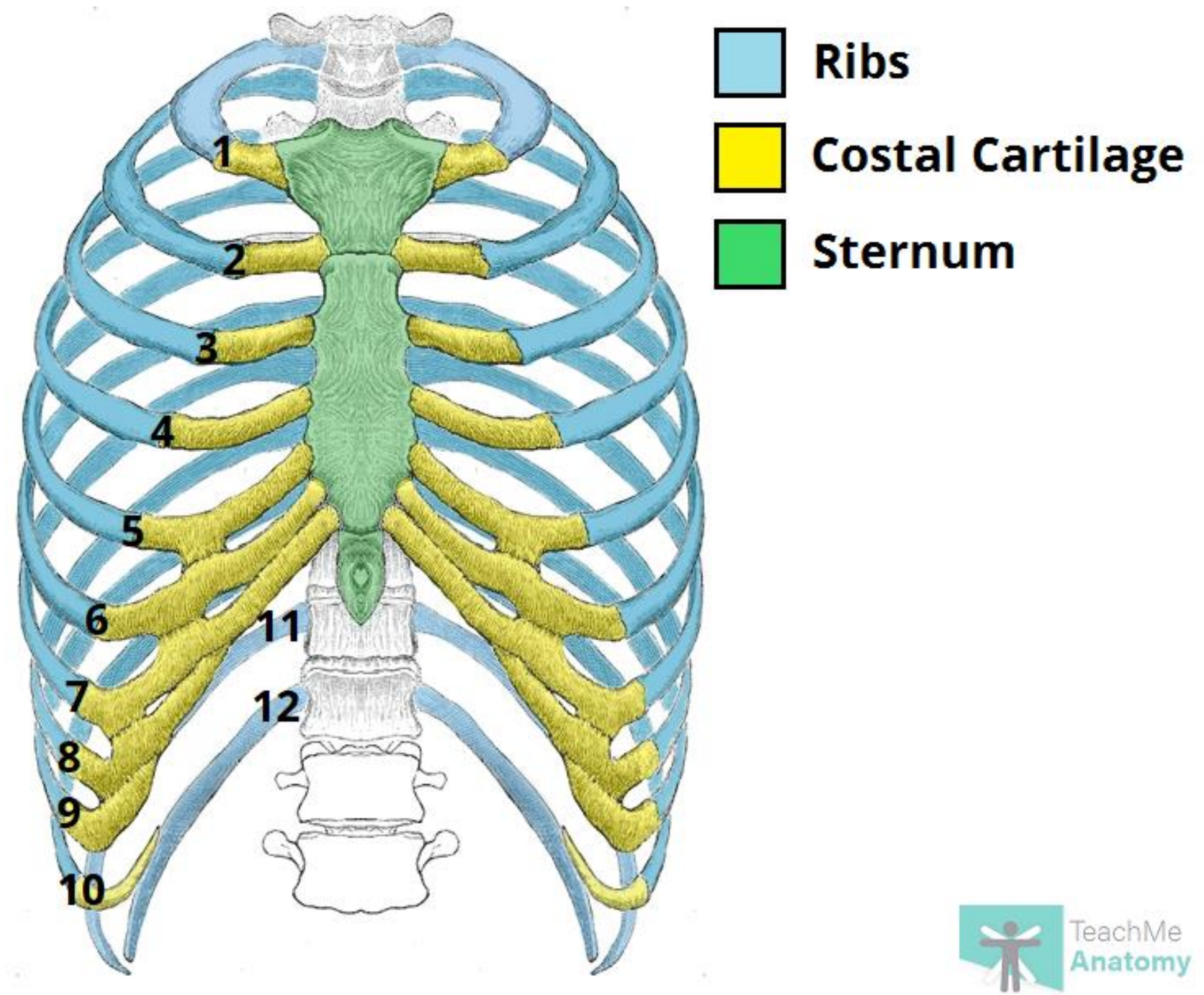
2- Contents of intercostal space

Intercostal muscles

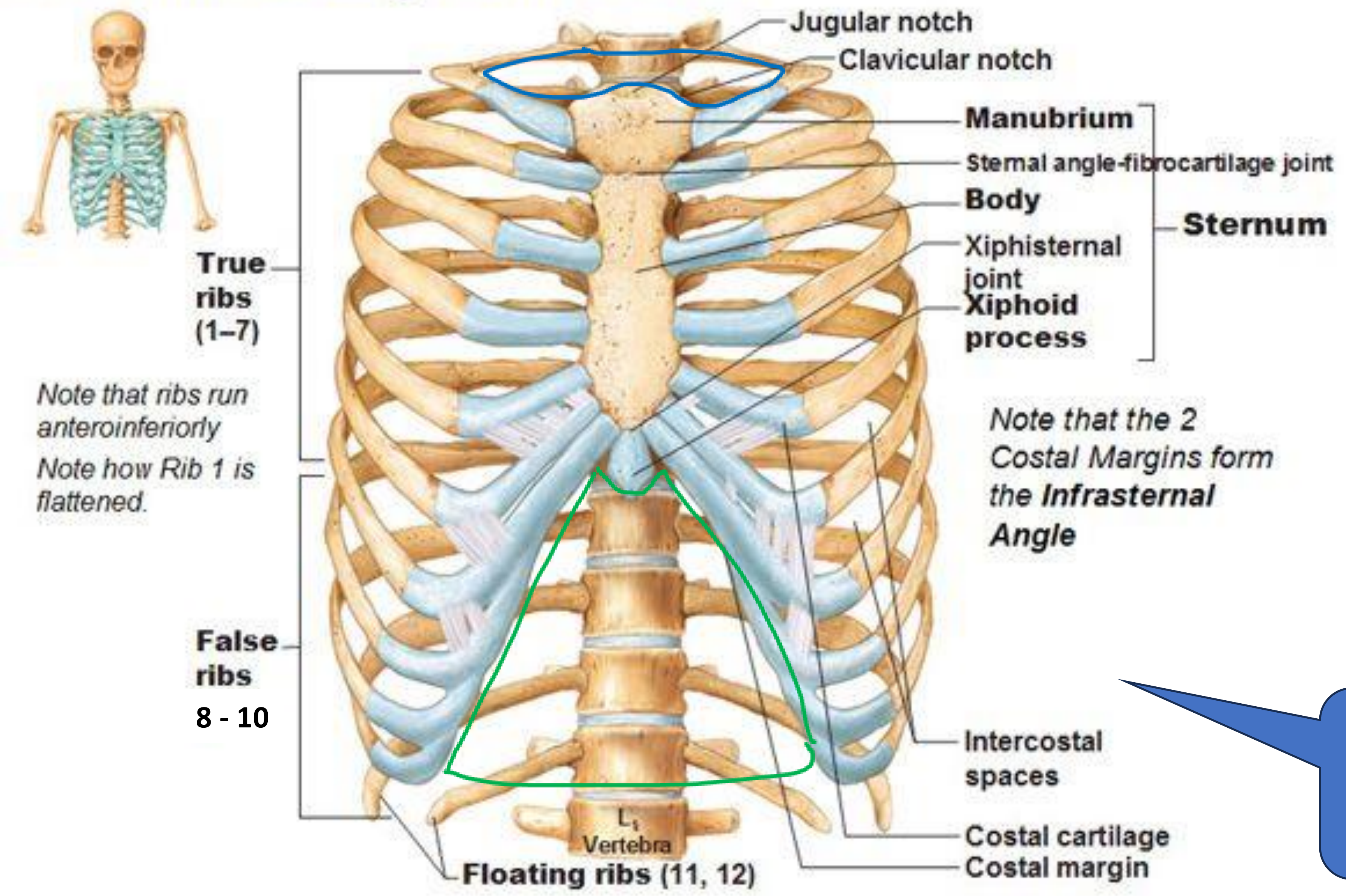
Intercostal nerves (typical & atypical)

Intercostal arteries (anterior & posterior)

Intercostal veins (anterior & posterior)



The Thoracic Cage: Anterior view



The **narrow upper opening** (**thoracic inlet**) Boundaries

The **lower wide opening** (**thoracic outlet).** Boundaries

The fused 7th,8th,9th & 10th costal cartilages. (costal margin).

Is the opening closed or opened?

Thoracic cage (skeleton of the thorax)

It is formed :

1- Posteriorly, by vertebral column (12 thoracic vertebrae)
2- laterally, on each side by ribs (12 ribs which enclose 11 intercostal spaces)
3- Anteriorly, by sternum & costal cartilages

NB: **The costal margin ❼ is the fused 7th,8th,9th & 10th costal cartilages.**

Thoracic cage has a **narrow upper opening** (**thoracic inlet**) may be called by the clinicians **thoracic outlet** as it transmits great vessels from thorax to the neck.

- **Boundaries of thoracic inlet:**

1- Posterior: by 1st thoracic vertebra
2- Laterally: by inner borders of 1st rib & its costal cartilage.
3- Anterior: by upper border of manubrium sterni

- **This opening contains**: trachea, esophagus, vessels, nerves & the apices of the pleurae & lungs (as the opening is oblique)

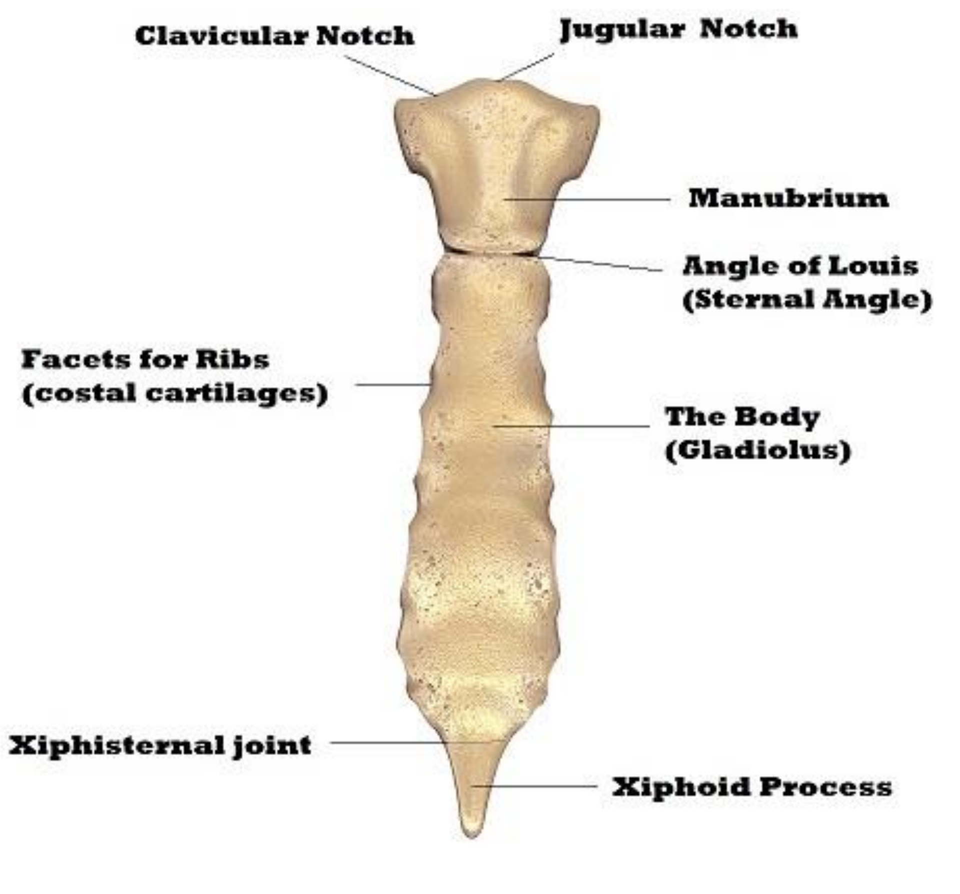
Thoracic cage has a **lower wide opening** (**thoracic outlet).**

- **Boundaries of thoracic outlet:**

1- Posteriorly: by 12th thoracic vertebra
2- Laterally: last 2 ribs & the costal margin (on each side)
3- Anteriorly: by xiphisternal joint & xiphoid process

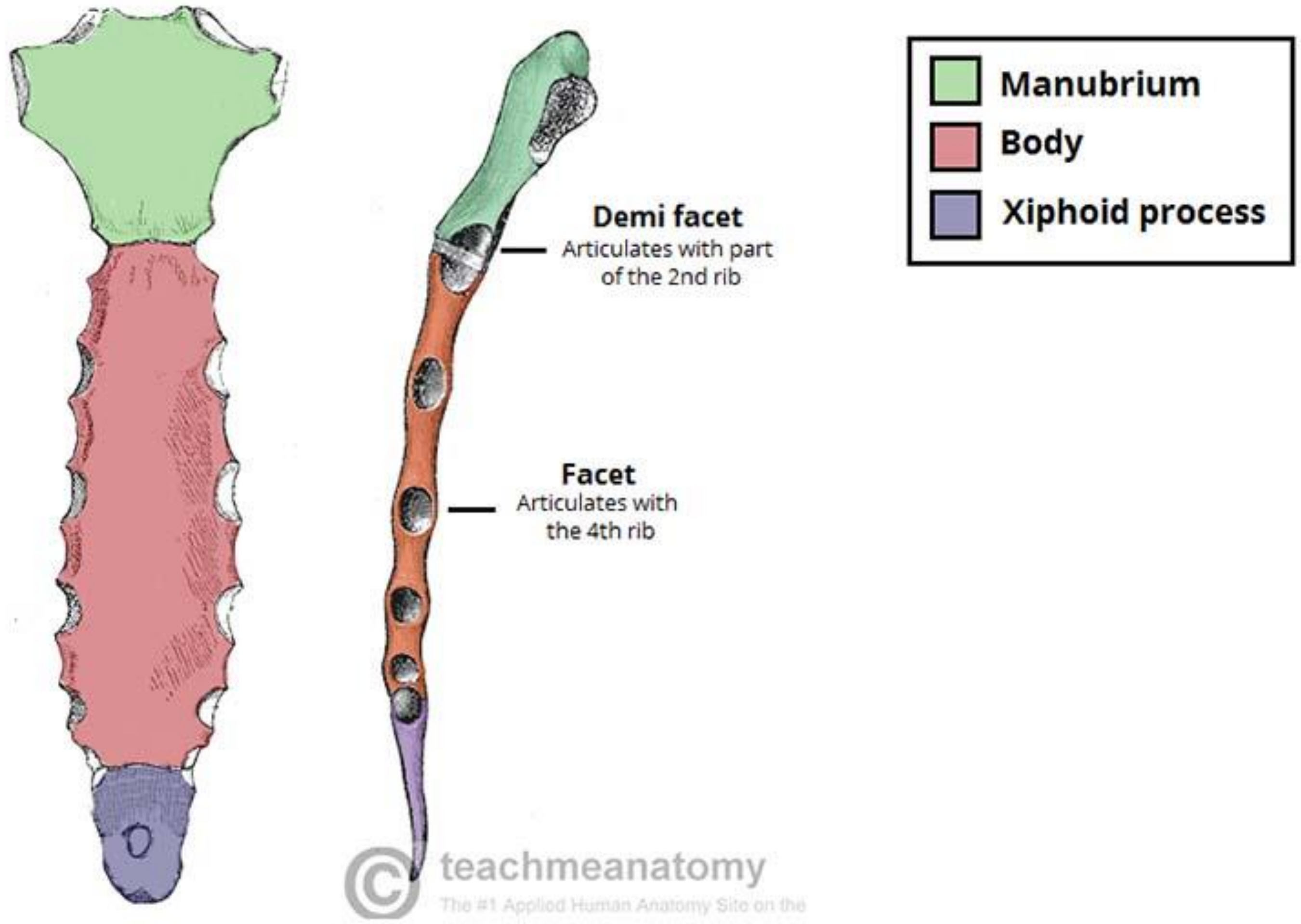
This opening is closed by the diaphragm.

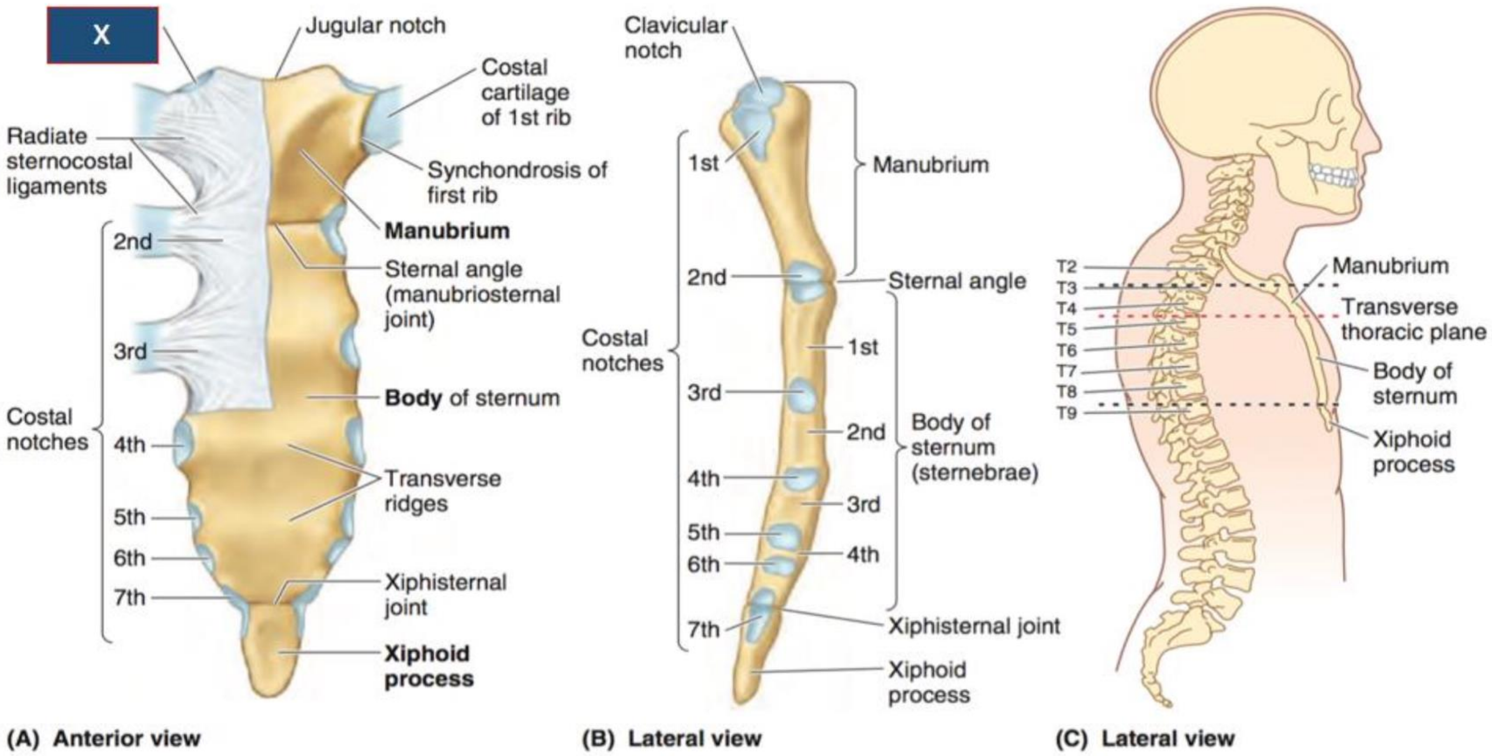
Sternum



Sternum: Anterior view

SciencePics/Shutterstock.com





Vertebral levels of sternal parts ?:

1- Upper border of manubrium sterni → between T2 & T3 vertebrae

2- Sternal angle → at the lower border of T4

3- Xiphisternal joint → at T9

Thoracic vertebra

They are 12 vertebrae ..

Typical thoracic vertebra (2nd – 8th):features are:

1- Body is small & heart shaped,

Body carries 2 costal facets (demifacets) for **head** of corresponding rib & rib below.

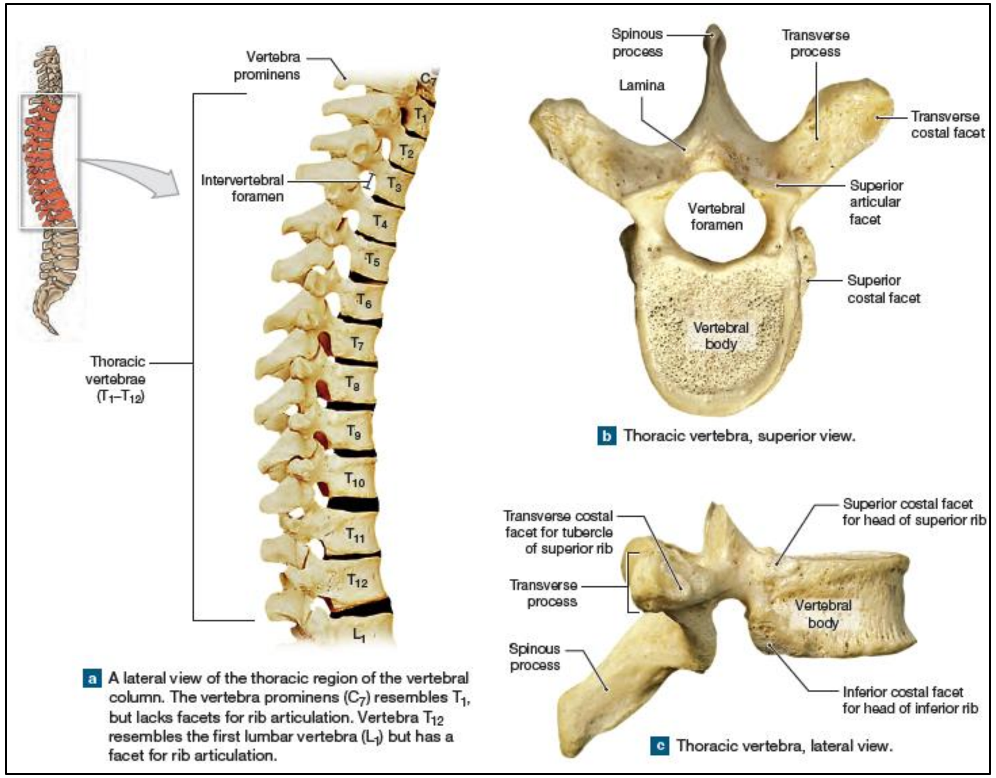
2- Transverse process contains facet for **tubercle** of corresponding rib.

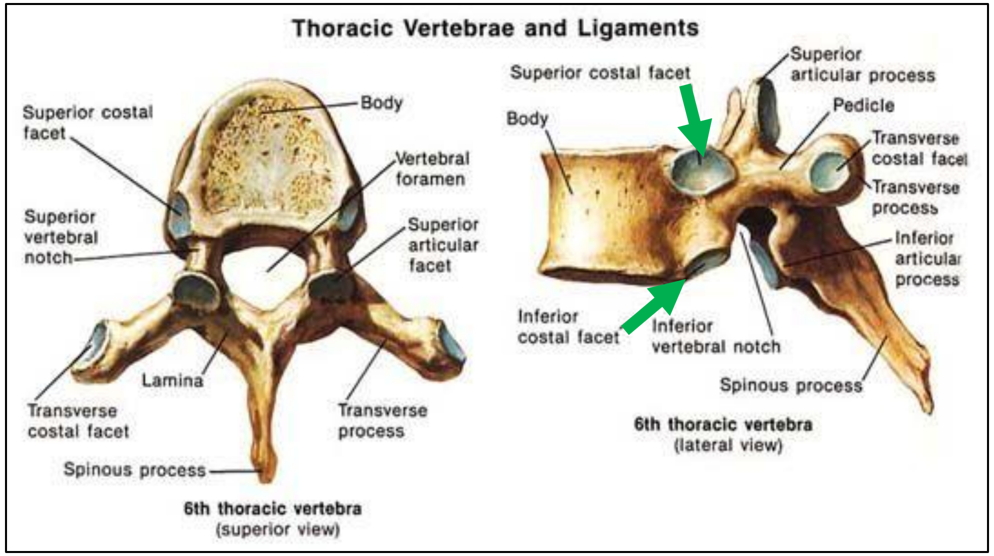
3- Spine is long & tapering, directed <u>downward</u>

4- **superior** articular processes are directed <u>backward</u>

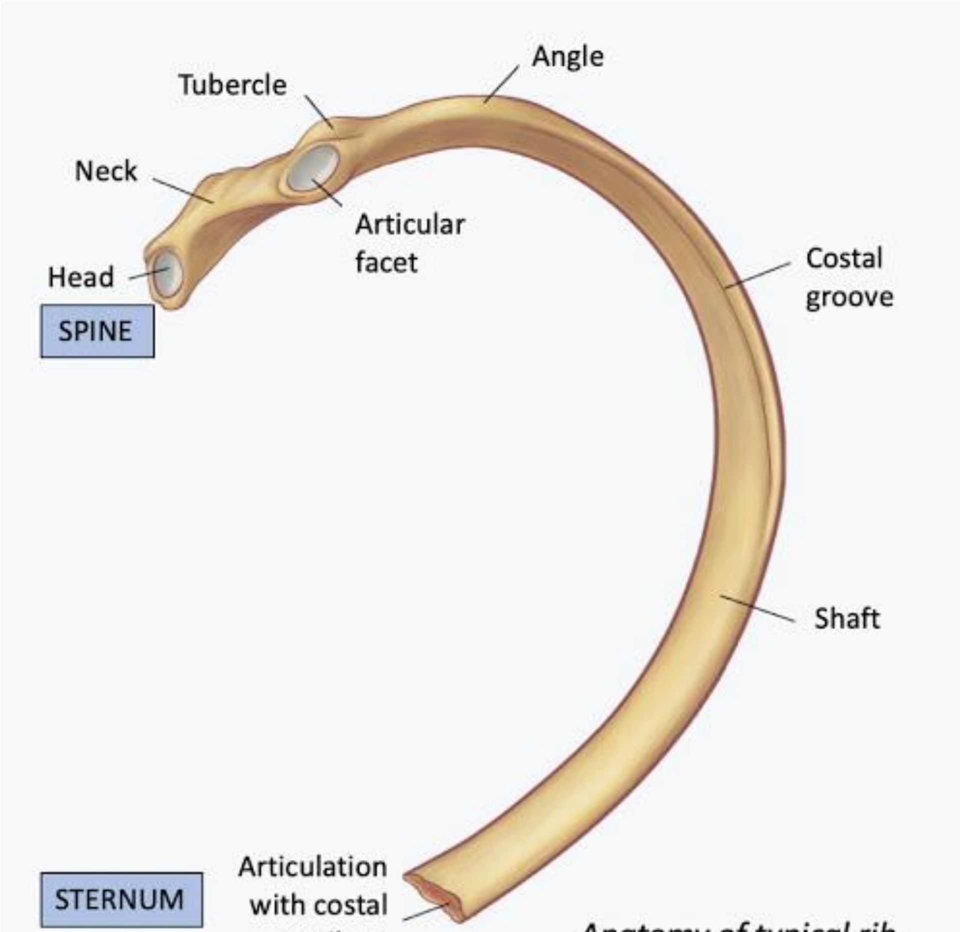
5- **inferior** articular processes are directed <u>foreword</u>

The **superior** and **inferior** articular facets are for articulation with vertebra above & below , respectively.





Rib



Anatomy of typical rib

Each rib has:

Head – neck- tubercle- angle- shaft- facet & costal groove.

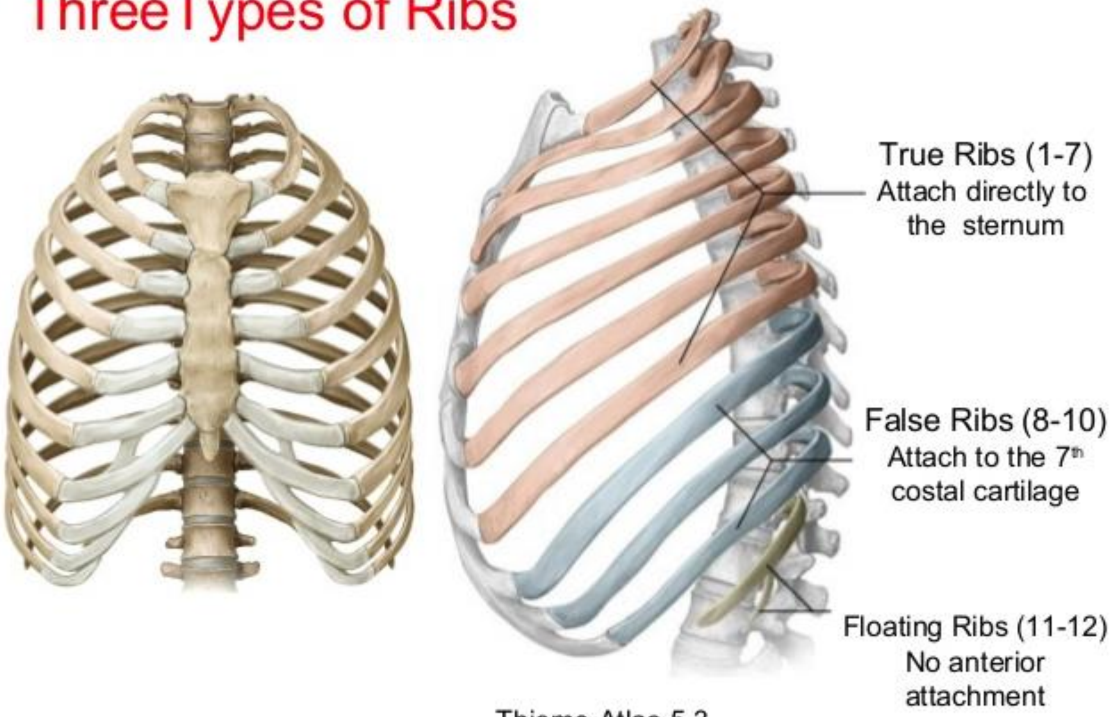
Subdivided into:

True ribs (1-7)

False ribs (8-10).

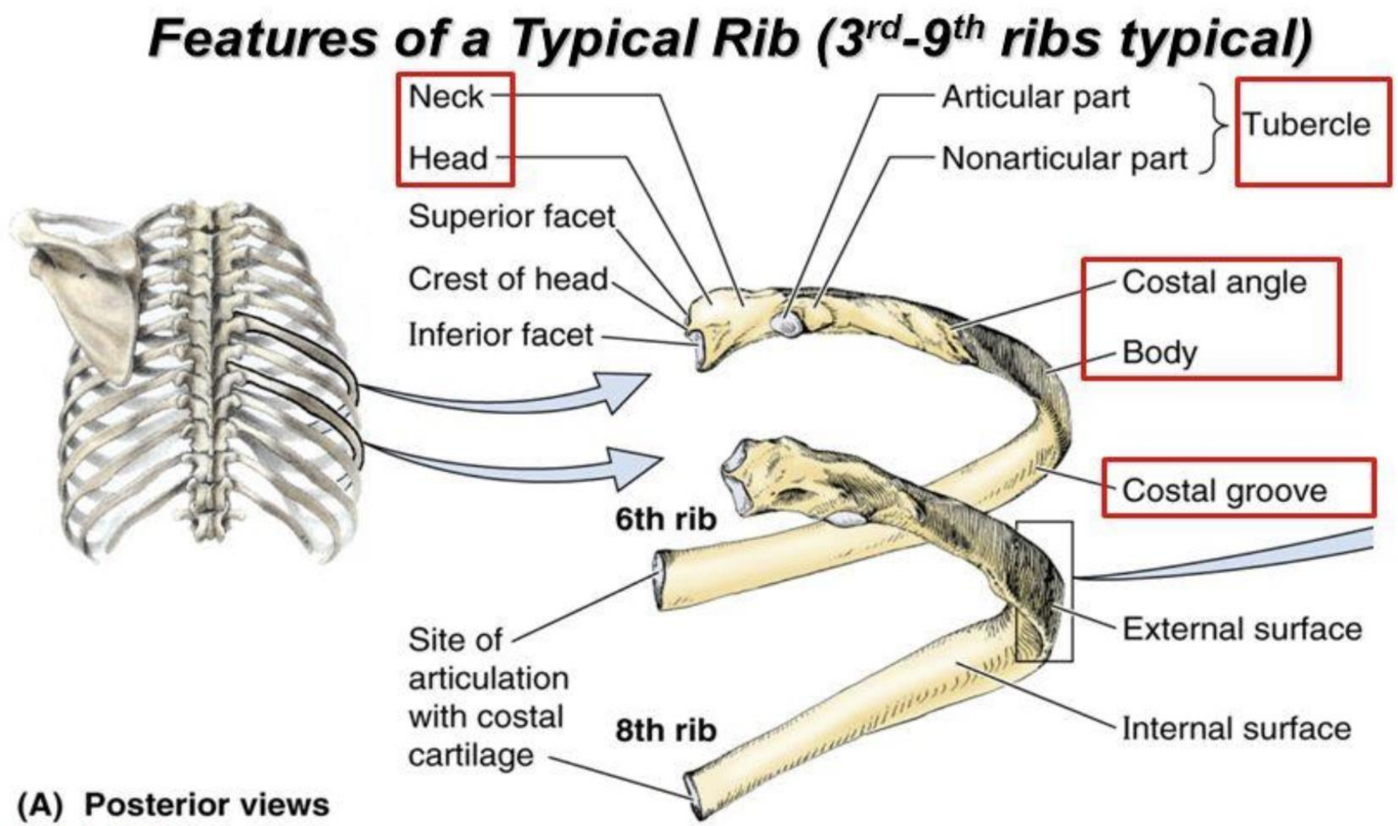
Floating ribs (11-12)

Three Types of Ribs



Thieme Atlas 5.3

Features of Typical Rib: (3rd – 9th):



(A) Posterior views

1- **The shaft** : is twisted

2- **Upper border**: rounded,

3- **Lower border**: sharp & contains subcostal groove (for intercostal VAN)

4- **Surfaces**: Outer convex surface, & inner concave surface,

5- **Anterior end**: is concave for costal cartilage ,

6- **Posterior end**: contains head, neck, & tubercle → the *<u>head</u>* contains 2 facets,

the *<u>tubercle</u>* contains single facet (synovial joints)

First rib:

it has upper & lower surfaces,
it has outer & inner borders.

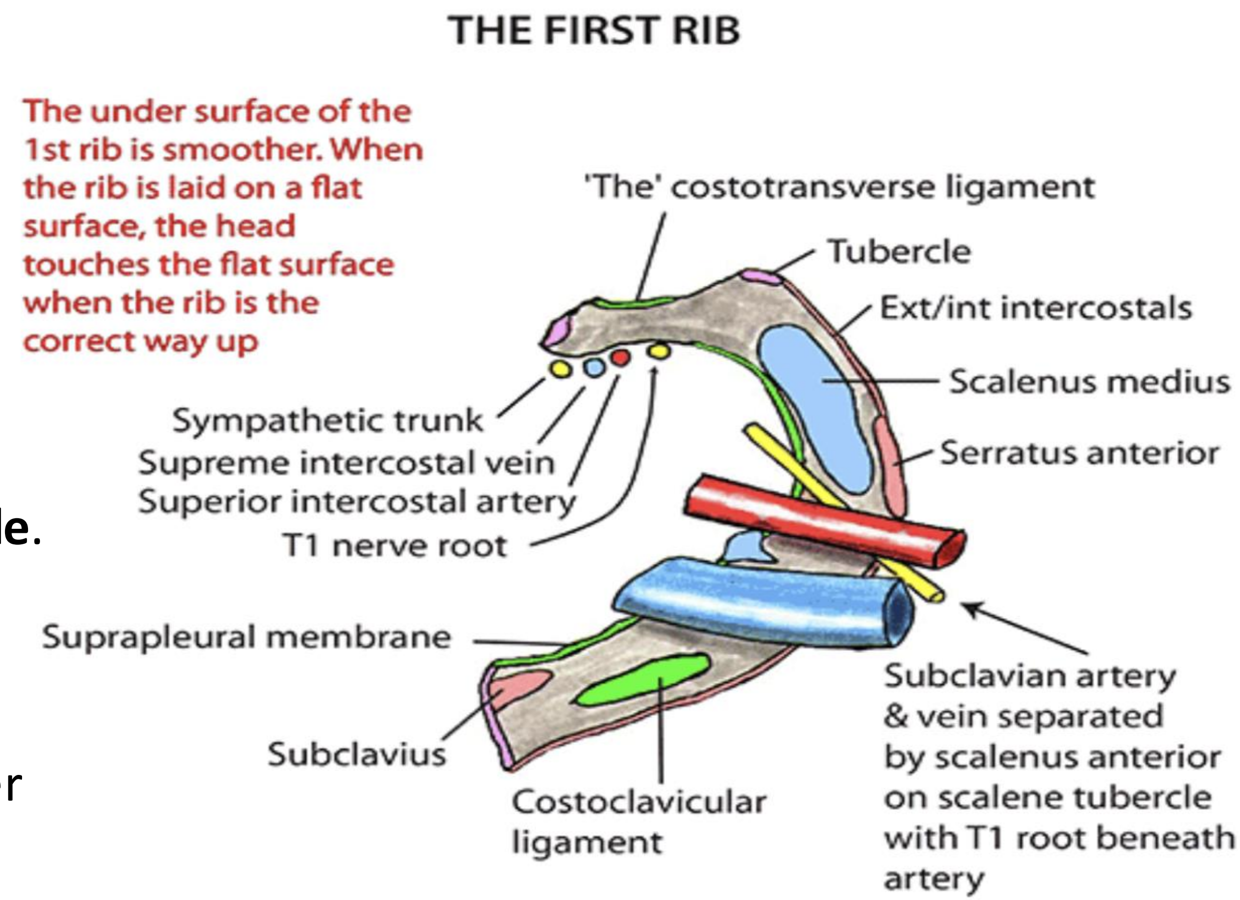
1- Posterior end: formed of
head (contains single rounded facet), **neck**, & **tubercle**.

2- Anterior end: concave, **NO twist in the shaft.**

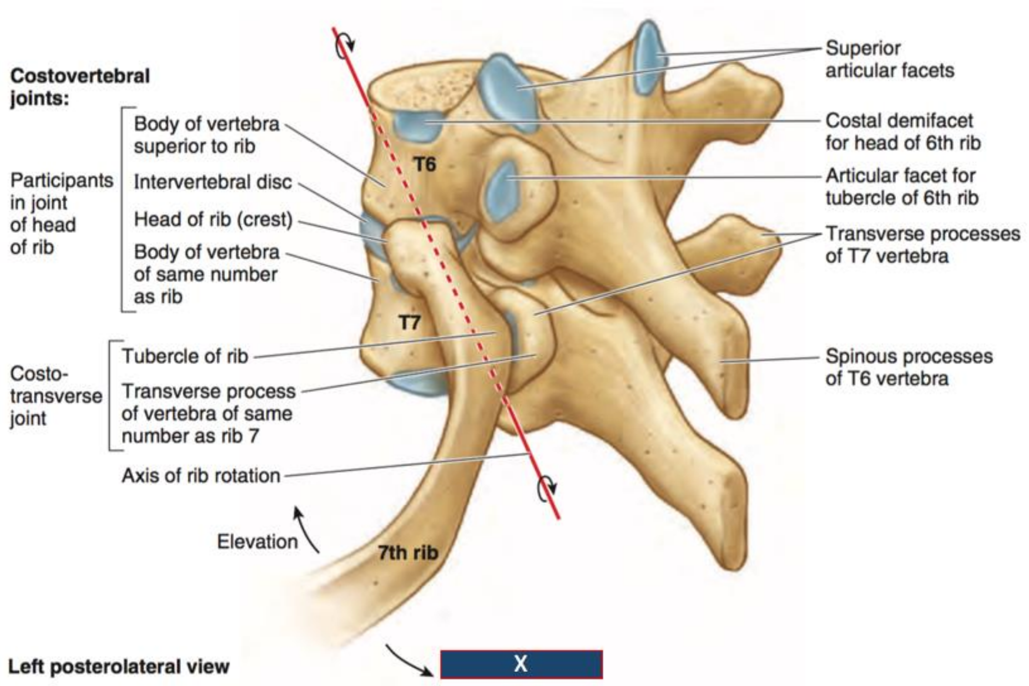
3- Borders: Outer convex border & inner concave border
(contains scalene tubercle)

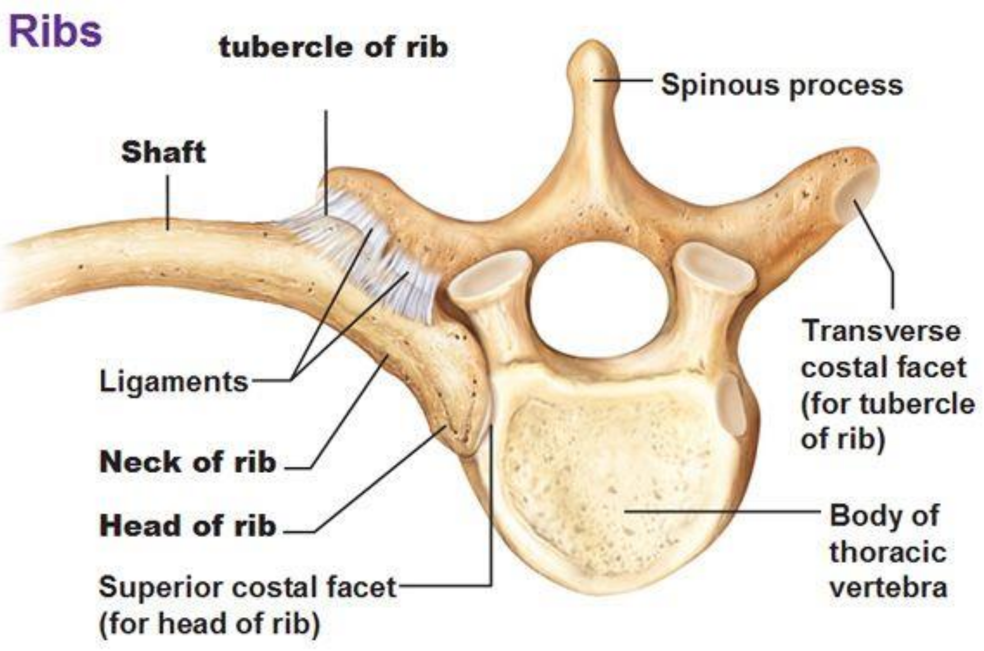
4- Lower surface: smooth.

5- Upper surface: rough & it contains two groove:
groove (in front of scalene tubercle) for subclavian vein.
& groove (behind scalene tubercle) for subclavian artery & lower trunk of brachial plexus



Rib & vertebrae





(c) Superior view of the articulation between a rib and a thoracic vertebra

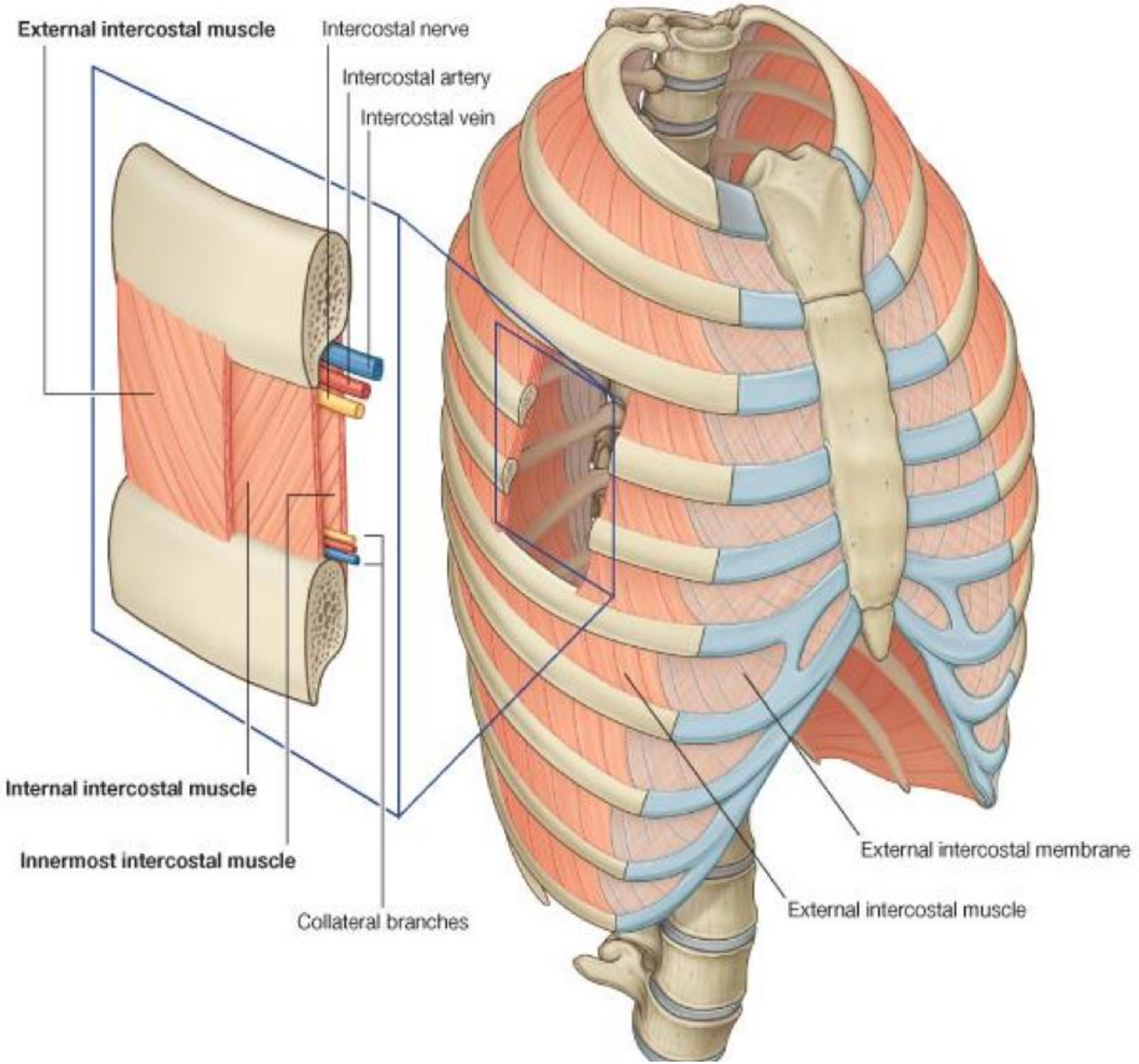
Each typical rib articulates with:

corresponding vertebra (same number) & vertebra above.

Each typical vertebra articulates with:

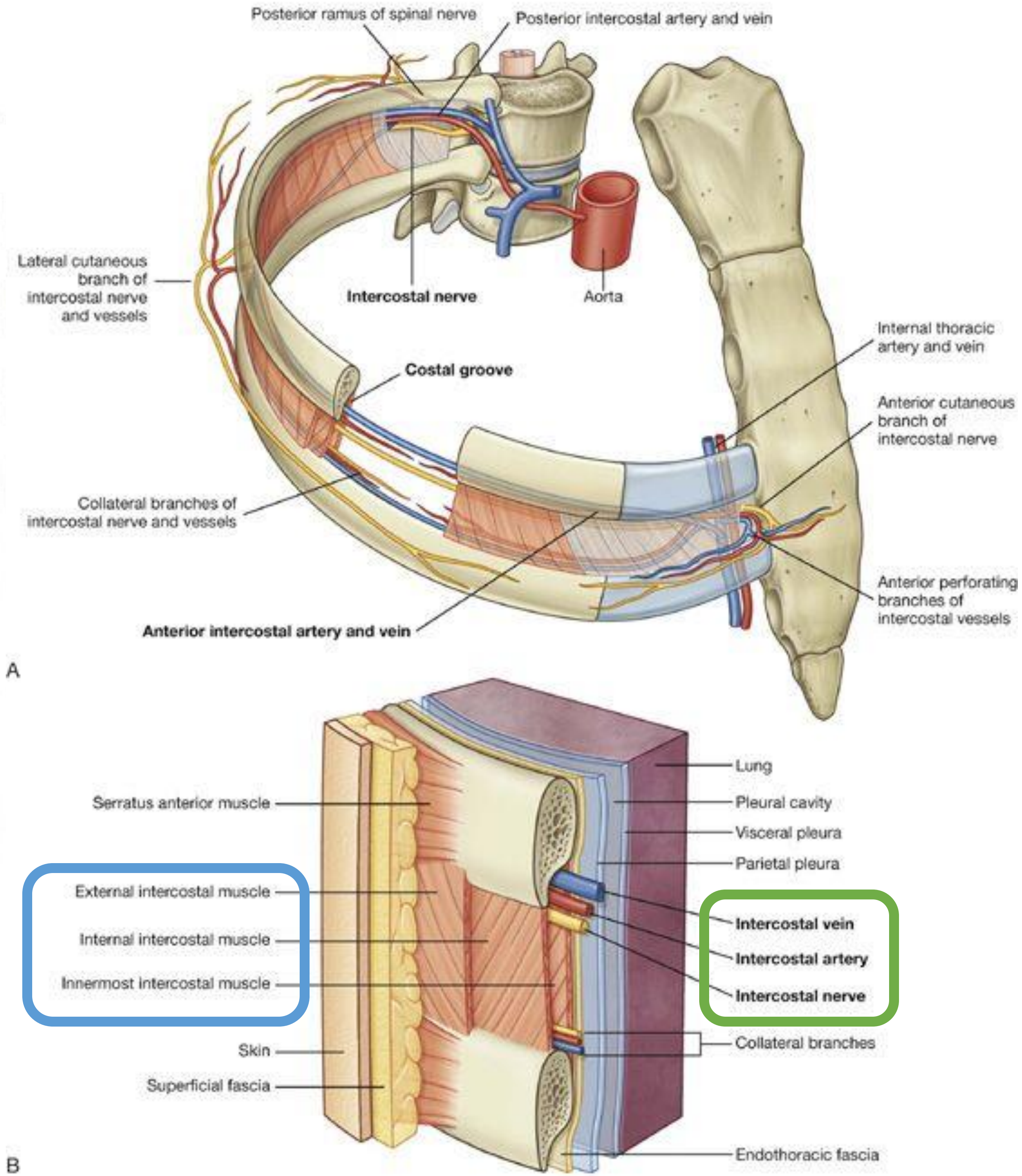
corresponding rib (same number) & rib below.

The intercostal spaces:



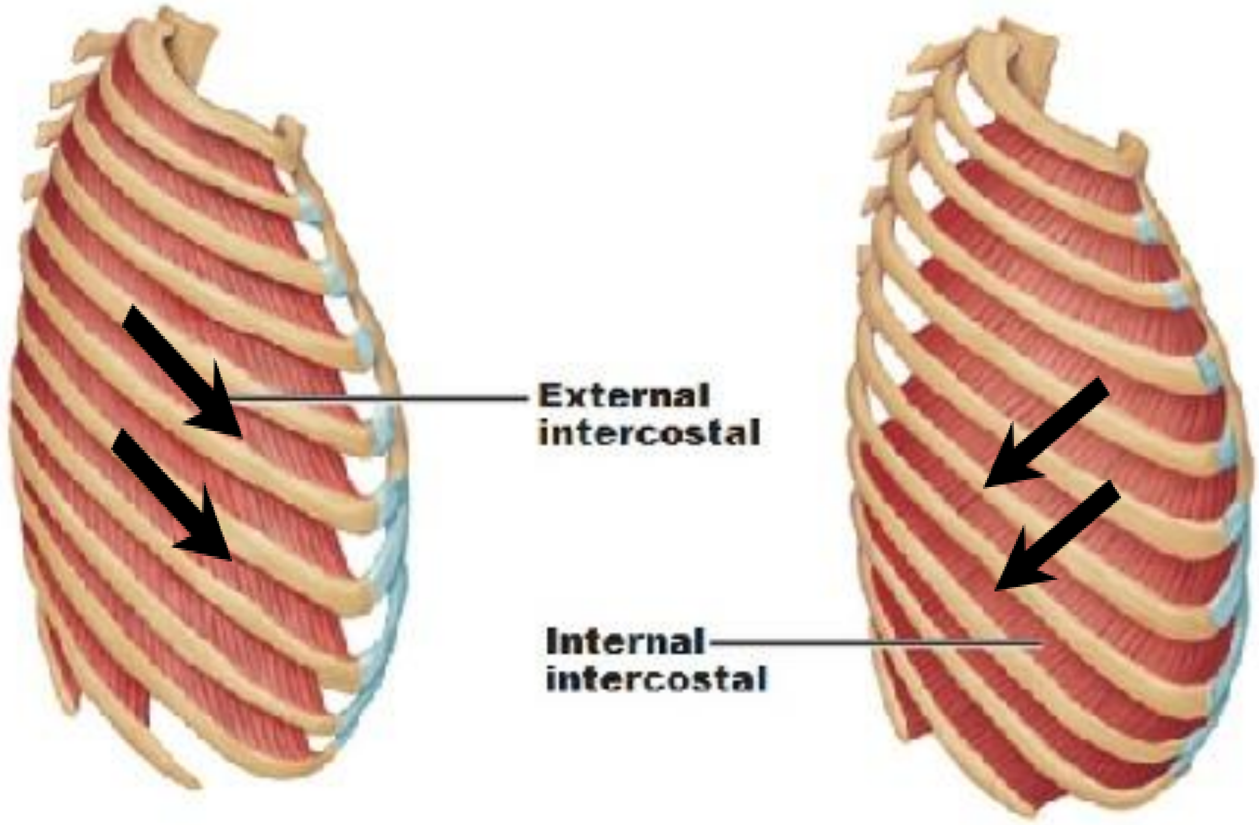
• Intercostal Spaces

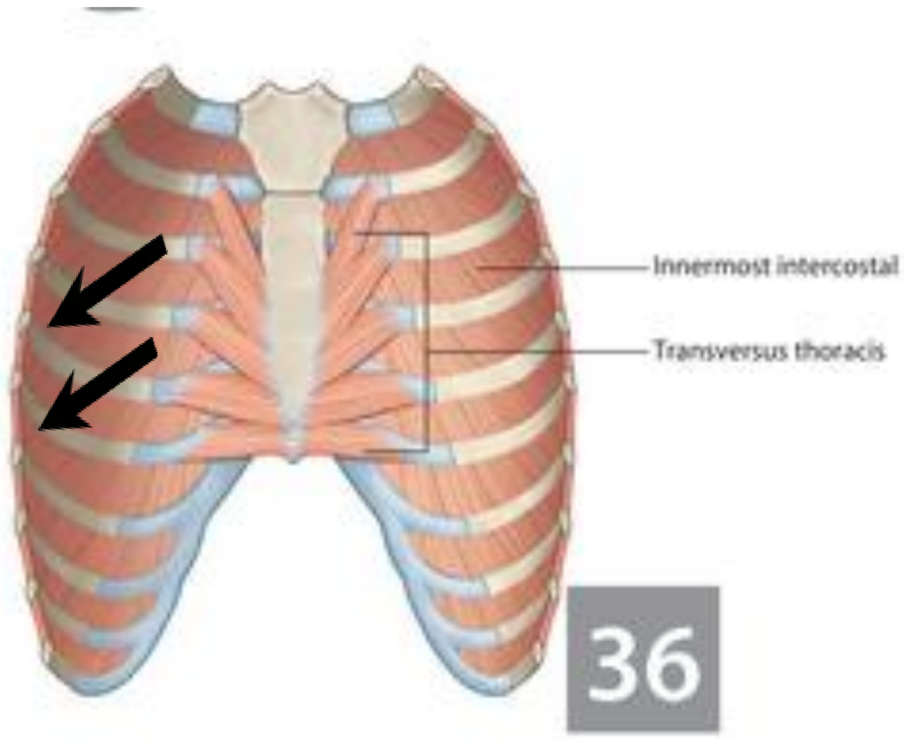
- There are **9** anterior and **11** posterior
- Each space contains:
- **1- Intercostal muscles**: (External, Internal and transversus thoracicus)
- **2- An Intercostal nerve.**
- **3- Intercostal vessels:**
- **a. Intercostal arteries** (Anterior & Posterior)
- **b. Intercostal veins** (Anterior & Posterior).



Direction of intercostal muscle fibers:

The fibers are directed….





External ICM

Downward
Forwards

Internal ICM

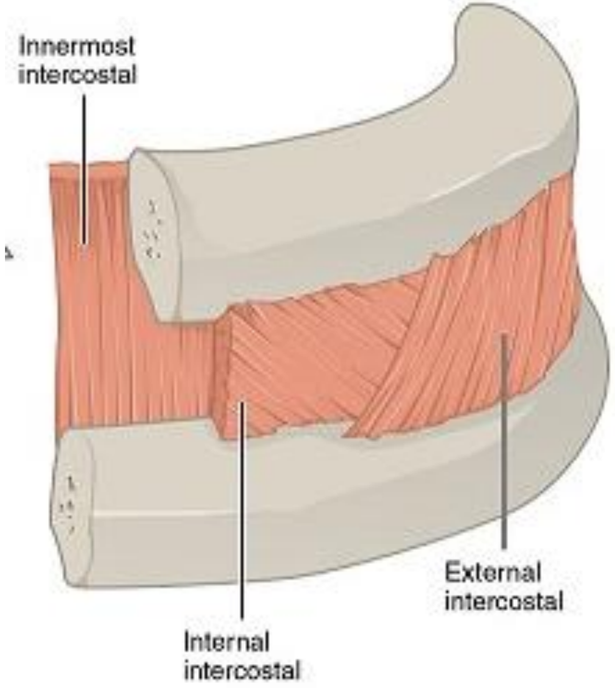
Downward
Backward

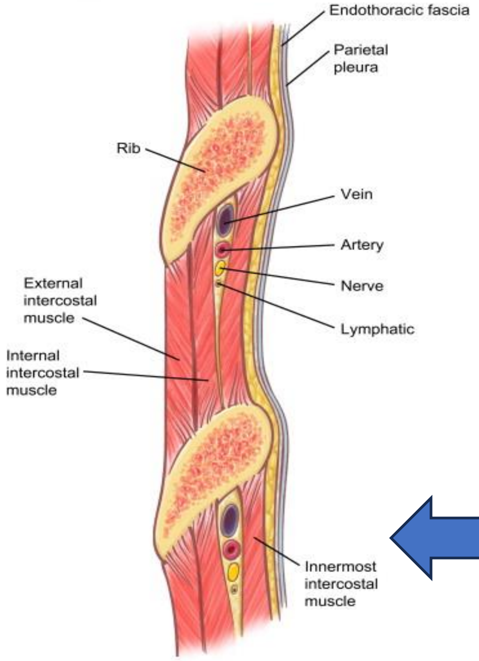
Innermost ICM

Downward
Backward

NERVE SUPPLY: of intercostal muscles by the corresponding intercostal nerves.

The intercostal muscles:





External intercostal muscle:

Origin: from lower border of rib above ❼
inserted into the upper border of the rib below
It starts posterior at the tubercle of the rib
At costochondral junction ❼it is replaced by
the anterior (external) intercostal membrane

Internal intercostal muscle:

Origin: from subcostal groove of rib above ❼
inserted into upper border of rib below
It begins anterior at the lateral margin of the sternum
At the angle of the ribs ❼ it is replaced by
the posterior (internal) intercostal membrane

Innermost intercostal muscle:

-It forms an incomplete layer (it is present only in the lateral part of thoracic wall❼it is absent in the anterior & posterior parts).

-The intercostal nerve & vessels are superficial to it
It may be considered as part of internal intercostal muscle (separated from it by intercostal nerve & vessels)

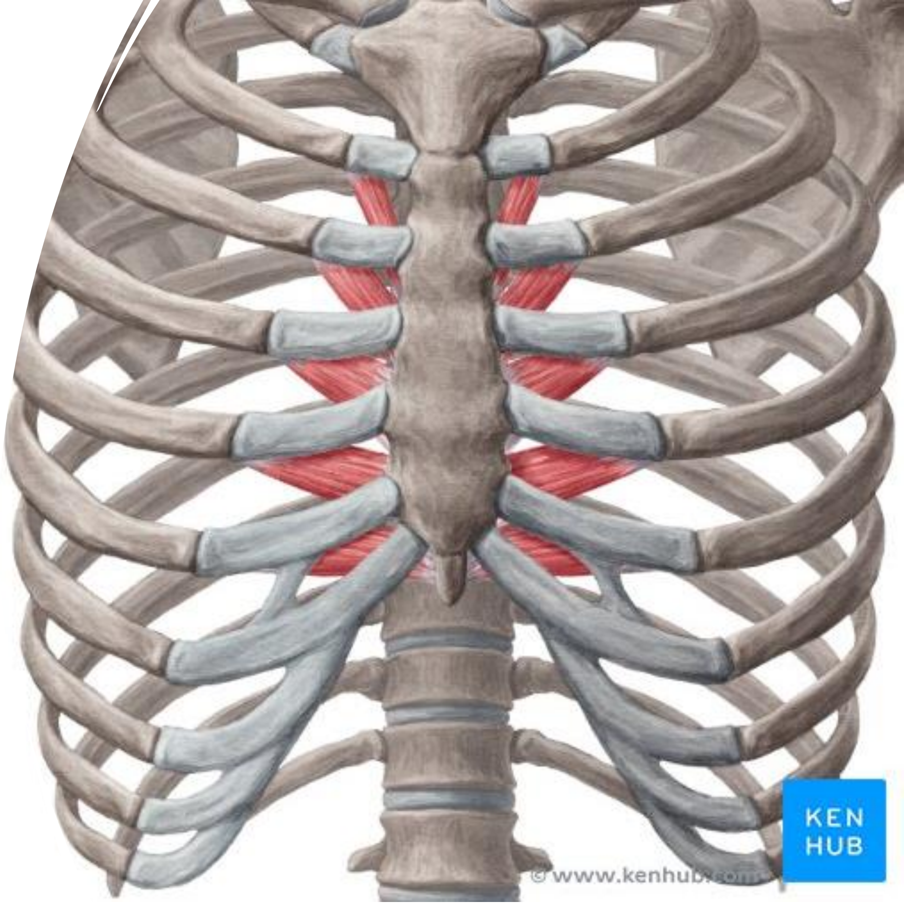
Transversus thoracis muscle (sternocostalis):

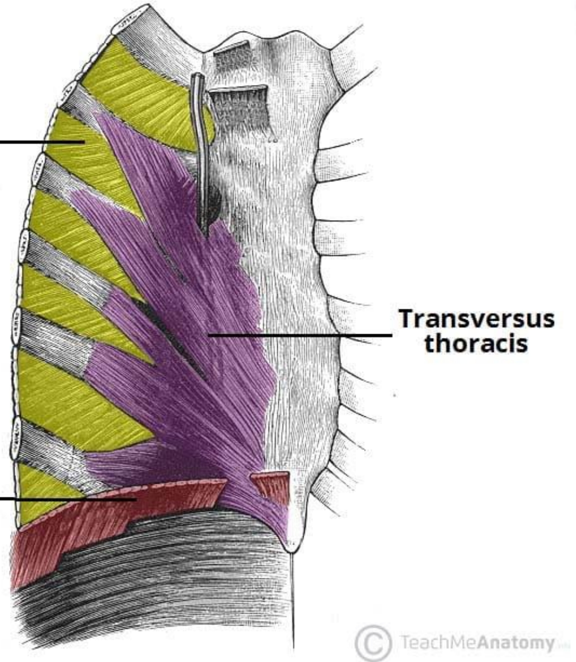
- Origin:

from posterior surface of lower ½ of ‘ sternum & xiphoid process

- Insertion:

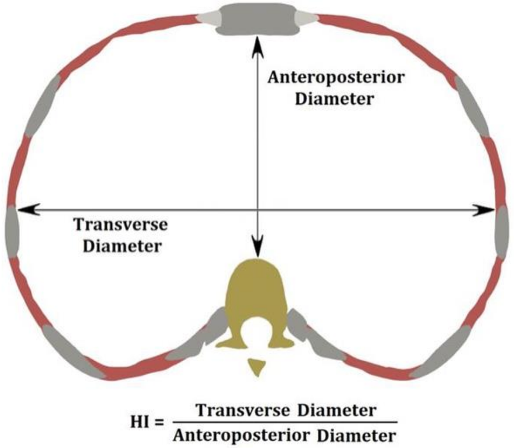
into posterior surfaces of the 2nd to 6th costal cartilages.



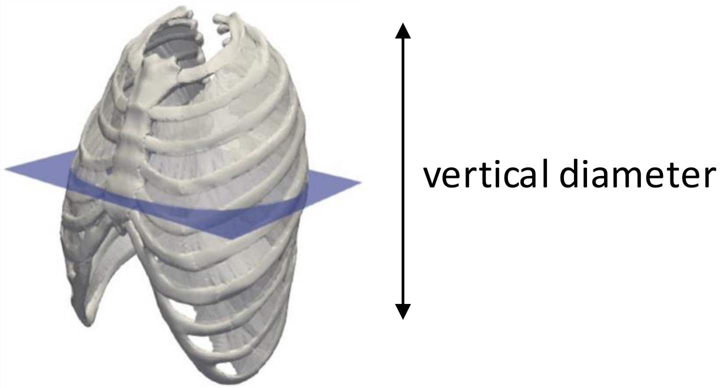


Breathing Mechanics

Breathing Mechanics



(A)



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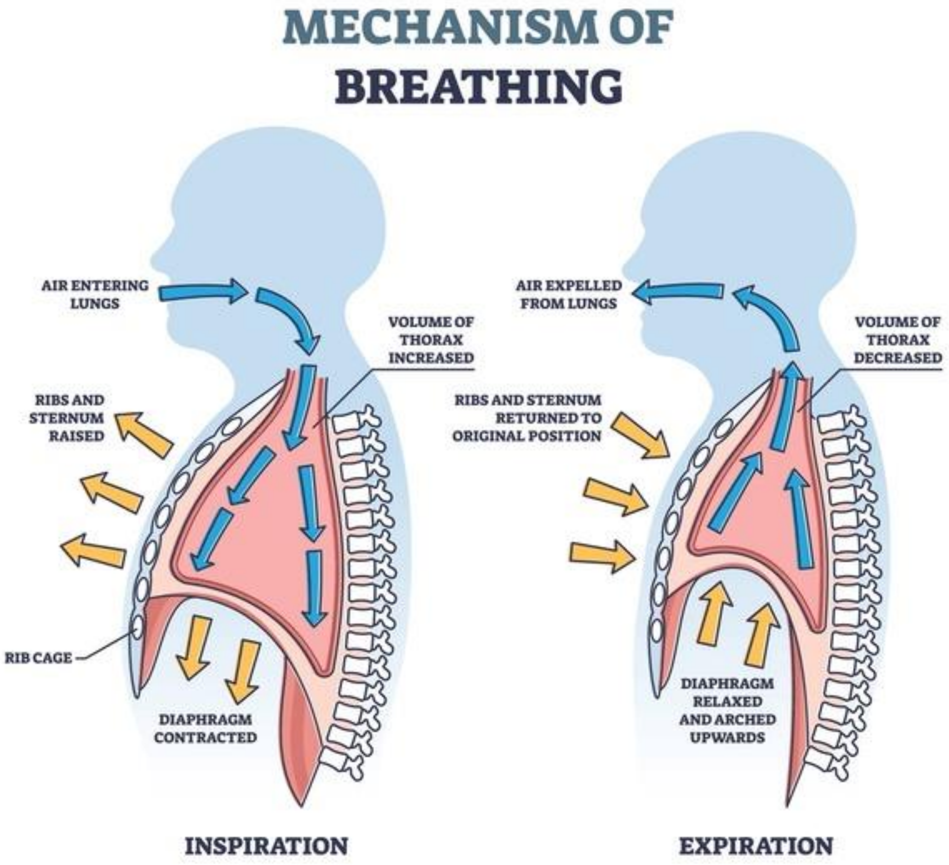


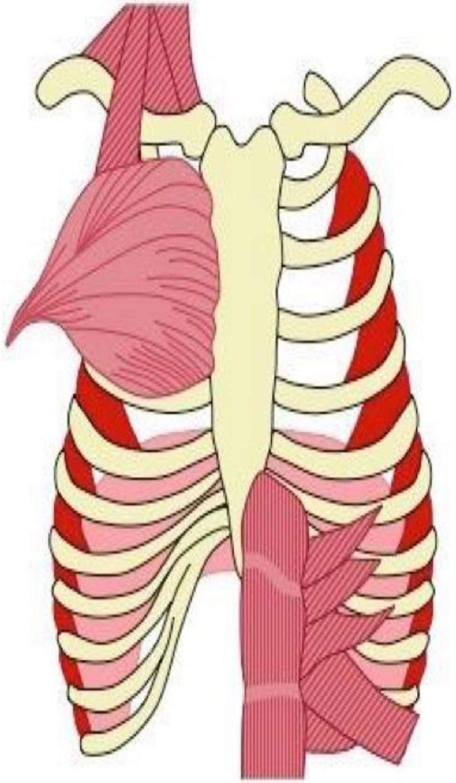
Muscles of Inspiration

Core Muscles

- External intercostals
  (contracts to elevate ribs)
- Diaphragm
  (contracts to expand thoracic cavity)

Accessory Muscles

- Sternocleidomastoid
  (contracts to elevate sternum)
- Pectoralis minor
  (contracts to pull ribs outwards)



Muscles of Expiration

Core Muscles

- Internal intercostals
  (contracts to pull ribs down)
- Diaphragm
  (relaxes to reduce thoracic cavity)

Accessory Muscles

- Abdominals
  (contracts to compress abdomen)
- Quadratus lumborum
  (contracts to pull ribs down)

Actions: of intercostal muscles→ are muscles of respiration

1- **Inspiration:** the 1st rib is **fixed** by contraction of scalene muscles, the other ribs are **raised up** toward the 1st rib by contraction of the external intercostal muscles

→ leads to increase the **transverse & anteroposterior diameters** of thoracic cavity

NB: contraction of diaphragm (descends)→**increases vertical diameter of thoracic cavity.**

2- **Forced expiration:** the last rib is **fixed** by contraction of the abdominal muscles, other ribs are **lowered (pulled down)** toward 'last rib by contraction of internal intercostal muscles

→ leads to decrease in the **transverse & anteroposterior diameters** of thoracic cavity

→ forcing air out ….

This is helped by contraction of muscles of the anterior abdominal wall which fix the diaphragm up.

3- what about **Quit expiration:**

It's a passive process needs just relaxation of respiratory muscles

→ lungs will recoil passively ;)

Watch this video :

https://www.youtube.com/watch?v=6bkjJWBBnCo

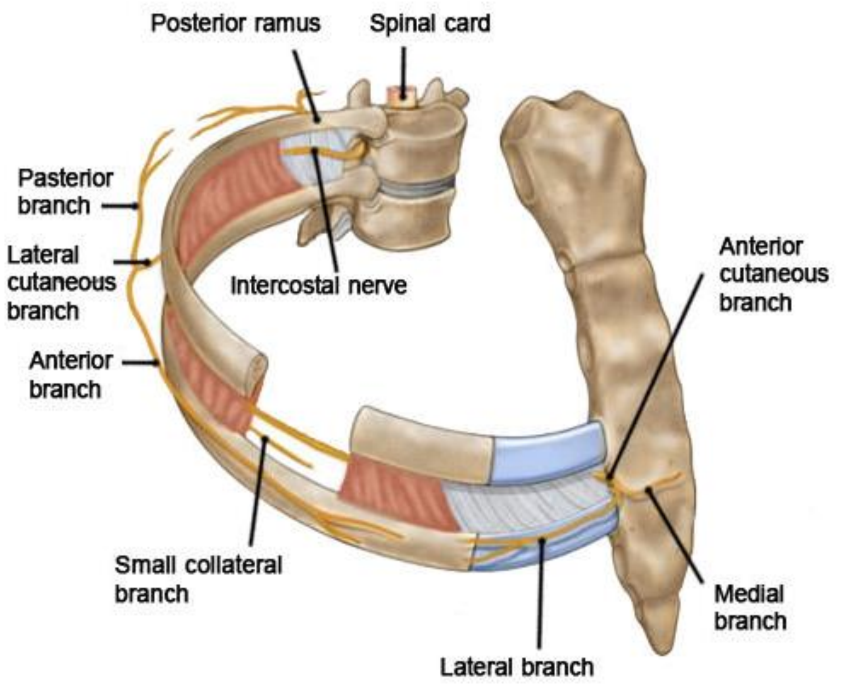
- Blood vessels
- Nerves
- Lymphatic drainage of thoracic wall

Intercostal nerves (11 pairs):

- They are the ventral rami of thoracic spinal nerves (from 1st to 11th).

12th ?! **Called subcostal nerve**

- Typical intercostal nerve (from 3rd to 6th):
- Origin: it is the anterior (ventral) ramus of thoracic spinal nerve
- Course: it runs anteriorly between internal & innermost intercostal muscles → it ends anterior as anterior cutaneous branch



- Branches: (Enumerate ?)

1- Branches to **sympathetic trunk**

2- **Collateral branch** → runs along the upper border of rib below

3- **Lateral cutaneous branch** → divides into anterior & posterior branches

4- **Anterior cutaneous branch** → divides into medial & lateral branches

5- **Muscular branches** → to supply intercostal muscles

6- **Sensory branches** → to supply parietal pleura

Atypical intercostal nerves:

- 1st intercostal nerve :

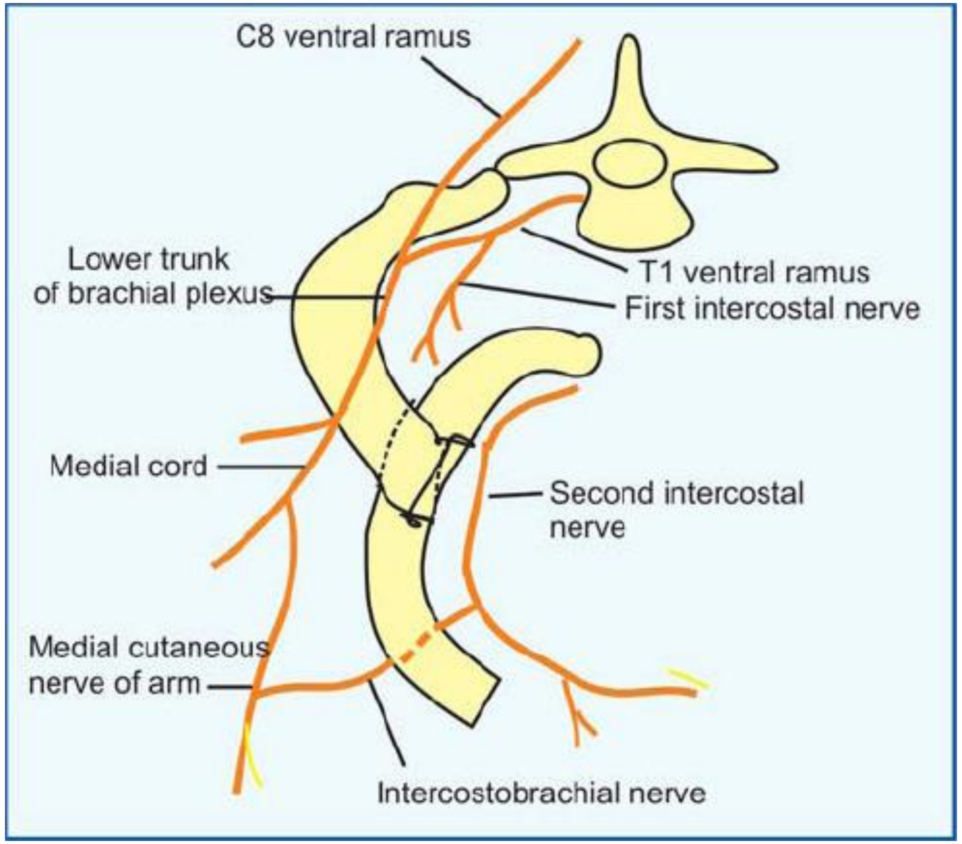
It is very small (as most of fibers of T1 share in brachial plexus), it has NO lateral & anterior cutaneous branches

- 2nd intercostal nerve:

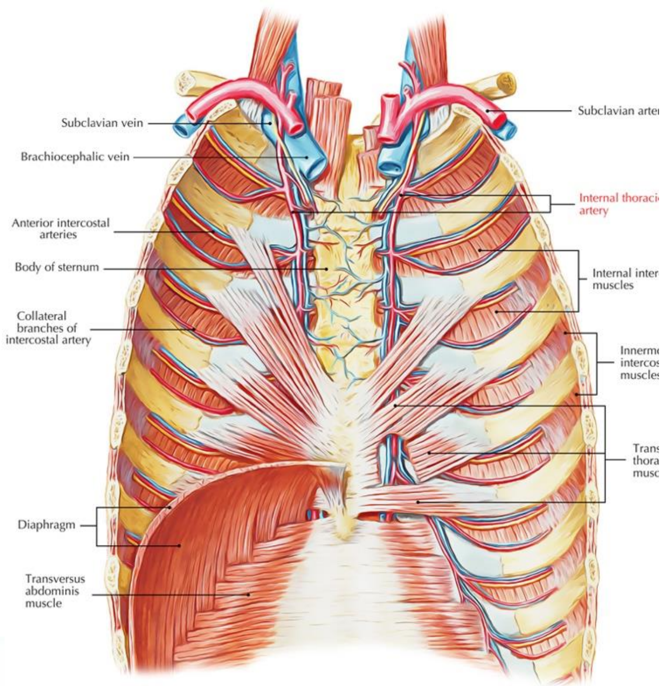
its lateral branch does not divide →
it forms intercosto-brachial nerve.
supplies skin over floor of axilla & medial side of arm

- Intercostal nerves (7th to 11th)

they enter the anterior abdominal wall
supplying its muscles, skin, & peritoneum



Internal thoracic artery



is a branch from subclavian artery.

Branches:

1- Anterior intercostal arteries (1st to 6th),

2- Perforating branches to breast

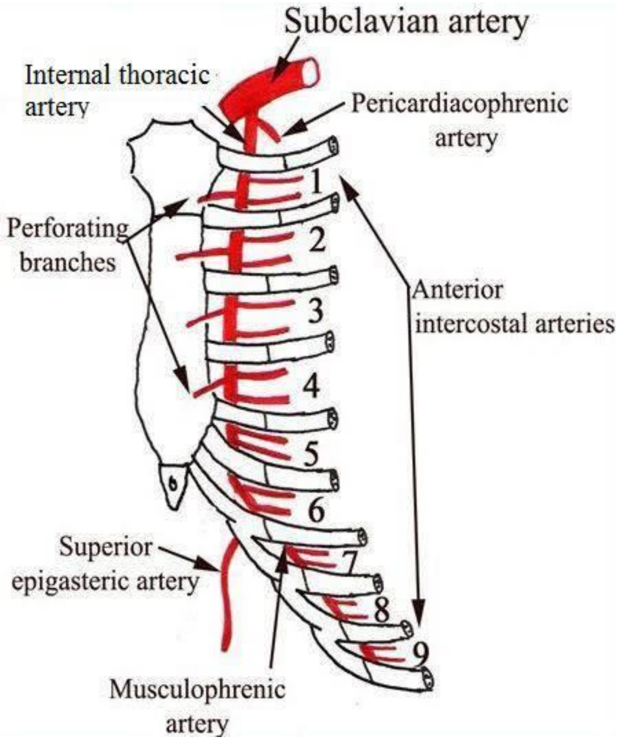
3- Branches to thymus gland,

4- Mediastinal branches → to mediastinum

5- Pericardiacophrenic artery

6- Musculophrenic artery

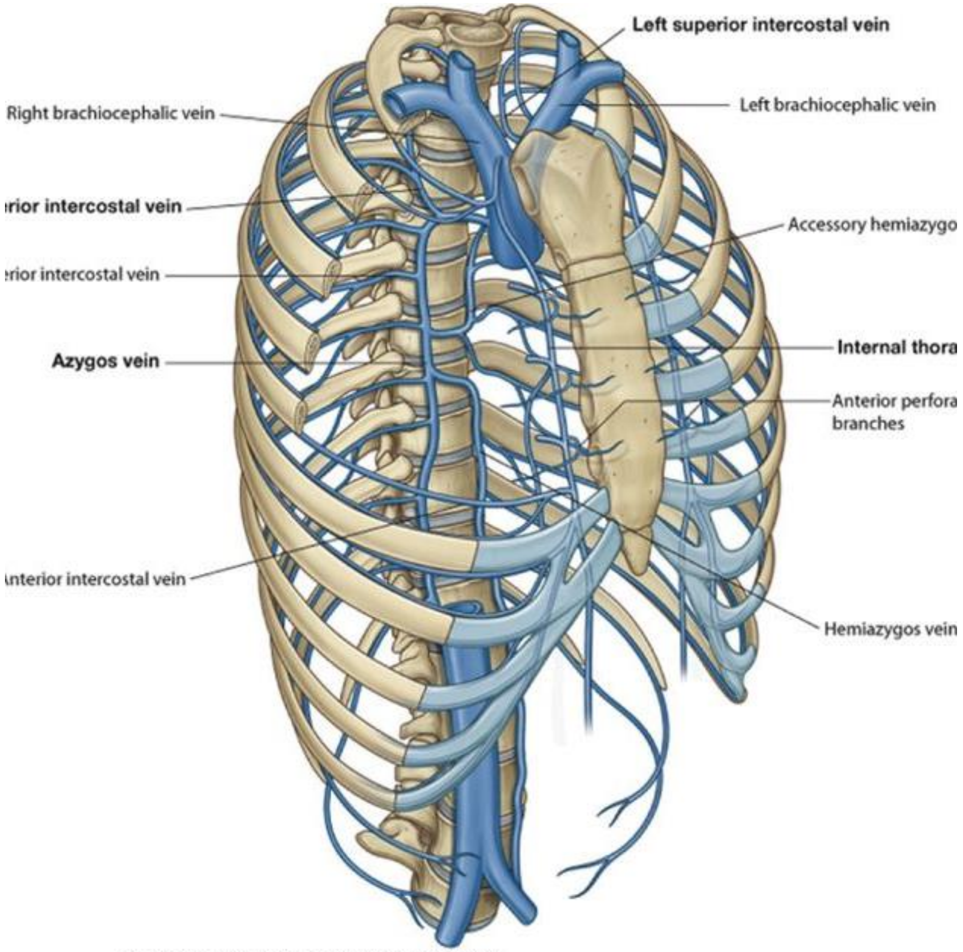
7- Superior epigastric artery



Internal thoracic vein

It is **formed** at 3rd intercostal space (by union of vena comitants)

it **ends** in the brachiocephalic vein

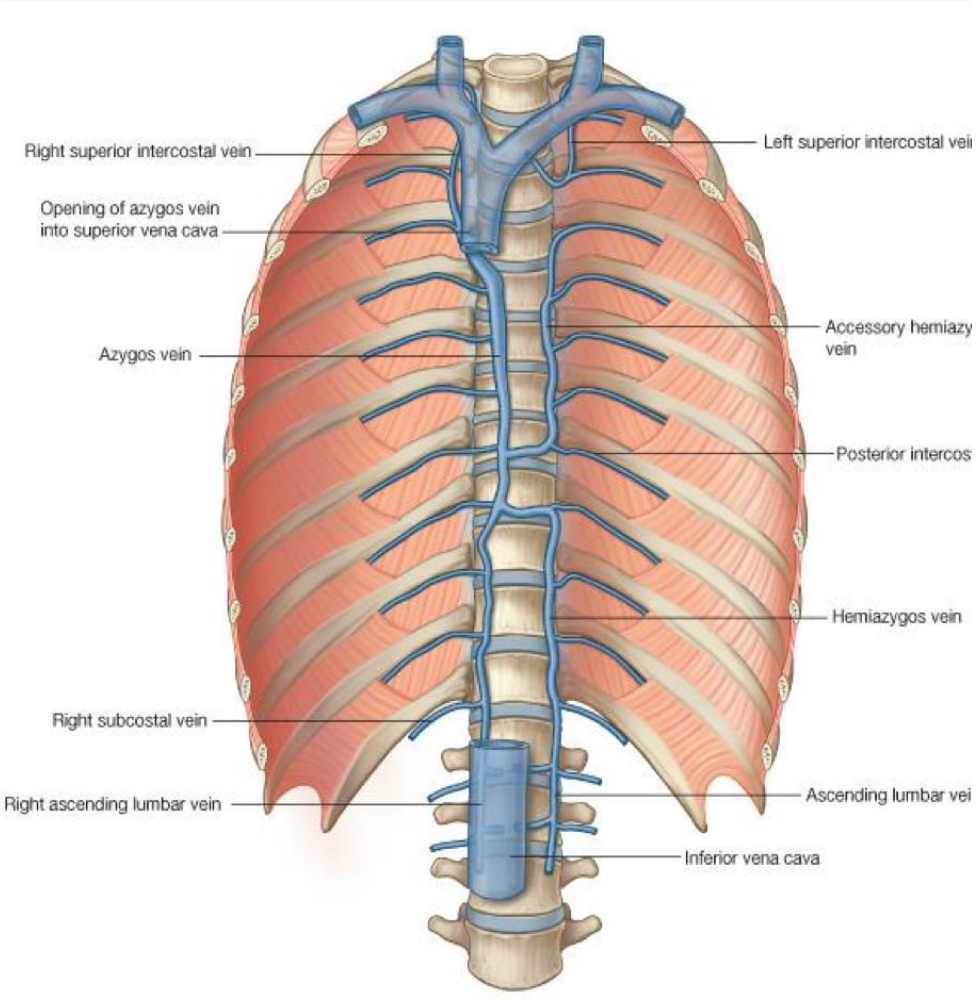


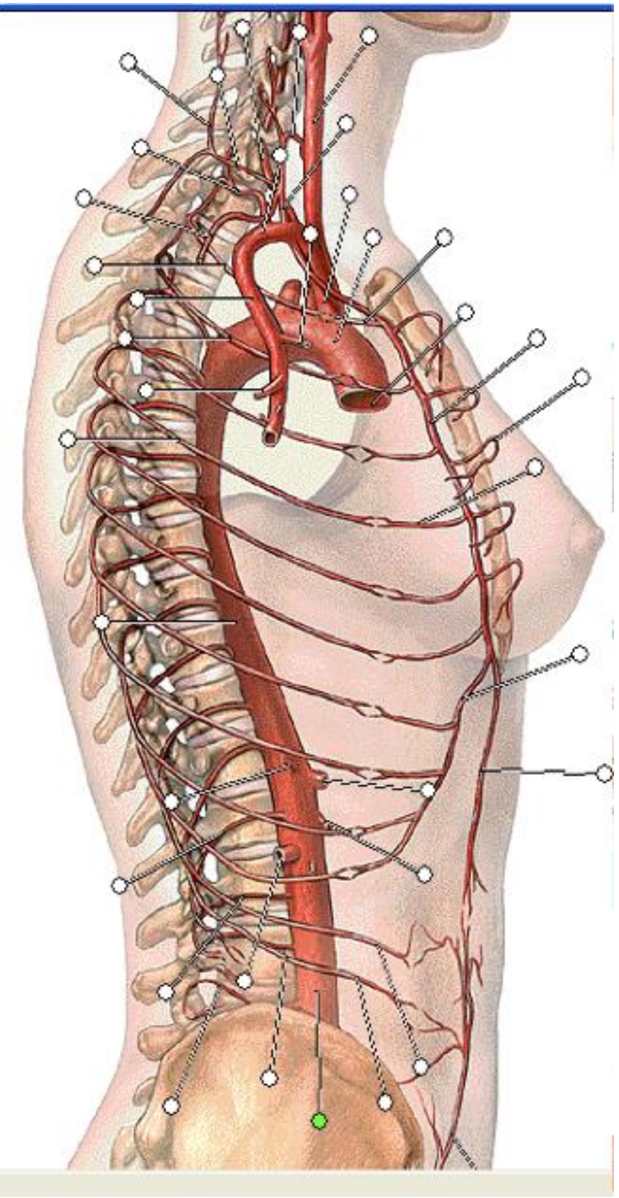
Drake: Gray's Anatomy for Students, 2nd Edition.

The azygos vein

- It begins from IVC in the abdomen
  It ends in the SVC in the thorax

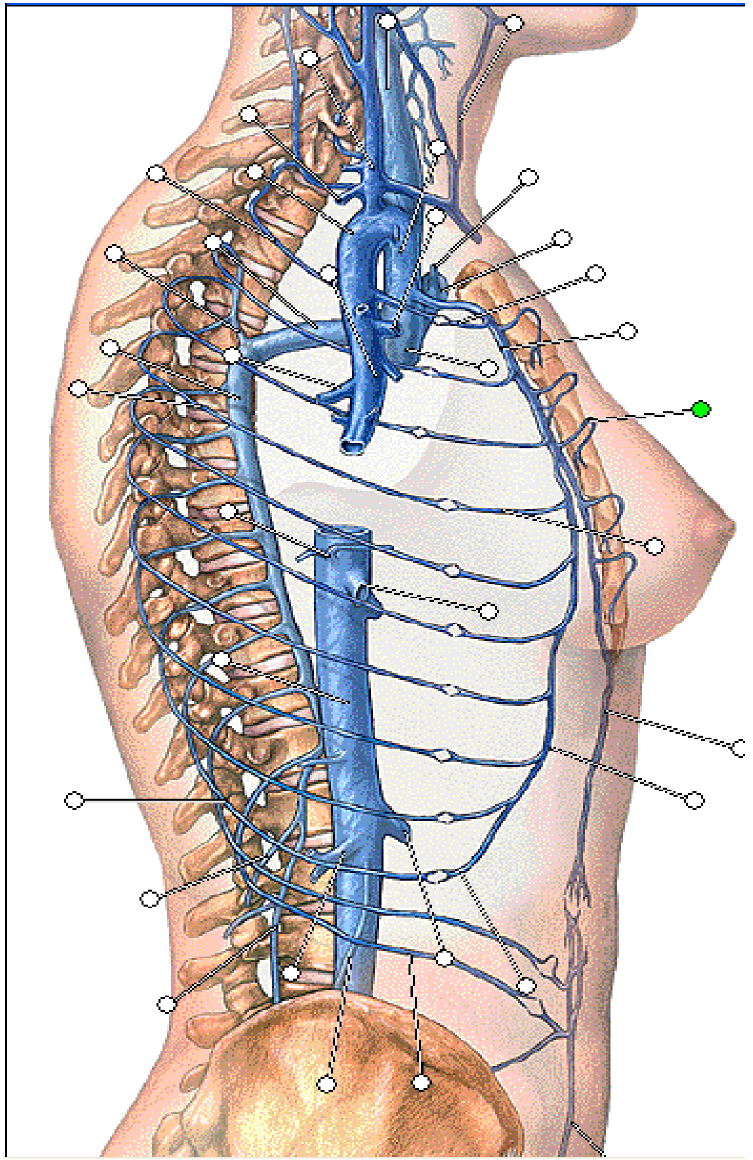
- From **LT side:** It receives hemiazygos veins
  From **RT side:** It receives RT posterior intercostal veins





Anterior Intercostal Arteries :
From internal thoracic artery
Posterior Intercostal Arteries:
From thoracic Aorta

Note the anastomoses between the anterior & posterior intercostal arteries



Anterior Intercostal Veins:
drain into internal thoracic vein

Posterior intercostal veins:
drain into the azygos vein (in the RT side) &
drain into the hemiazygos veins (in the LT side)

Lymphatic drainage of the thoracic wall:

A) Superficial lymphatics :

1- From anterior wall → into pectoral (anterior) group of axillary LN

2- From posterior wall → into subscapular (posterior) group of axillary LN.

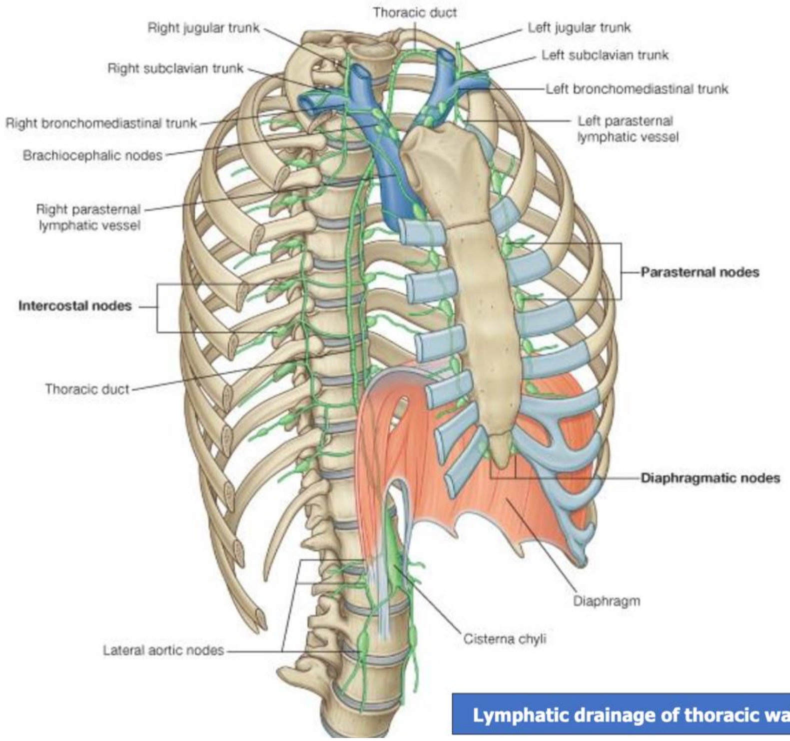
B) Deep lymphatics :

1- From anterior parts → into internal thoracic (parasternal) LN

2- From posterior parts → into posterior intercostal LN.

Lymph vessels from RT side of thoracic wall → end the RT lymphatic duct

Lymph vessels from LT side of thoracic wall → end the thoracic duct



Lymphatic drainage of thoracic wall

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Thank you